Vol 0, Chapter 12 — Time, Derived

Time as the flow of one geometry-forced operator; the double cover as spin-statistics; charge as internal; gravity as a one-input reduction

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Chapter 12 — Time, Derived

For everyone (the one-paragraph story)

Most physics *starts* with time — you draw a clock on the wall and let things move against it. BST doesn't have a wall to draw on. It has one geometry (D_IV^5) and one act: commitment — the substrate writes an irreversible record (Chapter 11, "The Operation"). Time turns out to be nothing more than *how far that writing has run*. There is no clock behind the clock: "when" is a measure of accumulated commitment. And because the writing only goes one way — you can't un-commit — time has a direction built in. That's the whole result: time is not assumed, it is derived, and everything usually *posited* about it (that it flows, that it has an arrow, that it pairs a real "tick" with an imaginary "circle") follows from the geometry.

For the student (the derivation, tiered honestly)

1. The flow [Derived — Tier 0]

The substrate Hilbert space is H^2(D_IV^5), the Bergman space of the bounded symmetric domain D_IV^5 = $SO_0(5,2)/[SO(5)\times SO(2)]$ (forced as the unique automorphism-invariant Born space, T754). The commitment dynamics is a one-parameter semigroup,

$$\rho_{\text{commit}}(\tau) = \exp(-\tau J/\hbar),$$

and time is the flow parameter τ — a reading of how far the semigroup has run, not an external axis.

2. The generator is forced [Structure-Derived]

J is the linear conformal Hamiltonian — the SO(2)-center weight of K = SO(5)×SO(2), the energy of the minimal representation, with a half-integer spectrum (ground weight $E_0 = 3/2$). It is emphatically not the quadratic Casimir: a Casimir is central (constant on each irrep), so it labels *which particle* and cannot move anything. Only the non-central linear J generates motion — and its linearity is exactly what makes $\exp(-iJt/\hbar)$ the ordinary Schrödinger equation. J is the time; the Casimir is the particle-label.

3. The arrow is spectrum-positivity [Derived]

J is bounded below (spec $J \geq E_0 > 0$), so $\exp(-\tau J)$ is a contraction semigroup defined only for $\tau \geq 0$ — the flow runs one way. That positivity is the arrow of time: not an added postulate, just the statement that the energy operator has a ground state. (The elementary tick is $\hbar$/energy; its numerical value $\approx 10^{-120}$ s is *Identified*, not Derived.)

4. Two faces [Derived; standard theorem]

Because J is self-adjoint and bounded below, $\exp(-zJ)$ is holomorphic on Re $z \geq 0$, with two boundary faces: the real-time tick $\exp(-\tau J)$ (a one-way half-line) and the imaginary-time circle $\exp(i\theta J)$. These are the Euclidean and Lorentzian faces of *one* generator, tied by Wick rotation. What is BST-specific is not that time has two faces — that is standard — but that the generator J is *forced by the geometry*.

Three things the chapter is careful NOT to overclaim (the honesty)

- The degree-2 cover is spin-statistics, not a new prediction. The imaginary circle is represented on a degree-2 cover, and on physical particles $\exp(2\pi iJ) = (-1)$^{#Rac} = (-1)^F in every dimension — it *is* fermion parity (fermions turn 720°, bosons don't). This is the geometric restatement of spin-statistics, a consistency the theory must and does satisfy. (An earlier draft called it "forced because n_C = 5 is odd — the core result"; that was retracted — the dimension-dependence cancels between constituents.)
- Charge is internal, not temporal. Electric charge is *not* a reading of the time-circle — decisively, by degeneracy (the muon and neutrino share a conformal weight $\Delta = 7/2$ but carry $Q = -1$ and 0, so charge is not a function of J at all). Charge lives in the internal sector; its fractional thirds are the fingerprint of the N_c = 3 color structure (the geometry forces the *number* 3 as the short-root multiplicity; it does not host the *group* SU(3) as an isometry).
- Gravity: the structure is derived, the value is one input. From the built Kostant cubic Dirac operator D (sign-indefinite; D^2 Laplace-type), the Einstein–Hilbert density falls out with a definite sign (the Sakharov route, F60/F63). But the *value* of Newton's G is one dimensionful input — the boundary curvature radius — exactly as general relativity takes G as an input. This is a theorem, not a shortfall: dimensionless content (integers, a geometry, ratios) cannot produce a length, so one dimensionful input is the floor for *any* theory, and BST is at the floor.

The reward (why this chapter earns its keep)

Asking “is G the boundary curvature?” turned into a genuine reduction. BST’s one dimensionful input is not the Planck length (which would be circular) but the electron mass m_e — measured by trapping a single electron, never touching G. From it: m_p/m_e = $6\pi^5$ (0.002%), the Planck scale (0.03%), and hence G to 0.06%, plus the electroweak scale v. The honest headline is not “we predict G” — *nothing* predicts a dimensionful constant from nothing — but BST reduces the dimensionful inputs from two (G and v) to one (m_e), and makes the survivor an atomic measurement rather than a gravitational one. (Note: the m_e anchor and the G prediction are one relation read two ways — G $\propto$ m_e^{-2}, so its error is exactly twice m_e’s — not two independent confirmations.)

Connections

- Ch 11 (The Operation) names time as the SO(2) common append-axis; this chapter derives the flow, arrow, and faces of that axis.
- Vol 4 (GR/Cosmology) carries the gravity thread (the 2→1 reduction; gravity-as-geometry).
- Vol 5 (QM) — the linear J *is* the Schrödinger generator (Measurement-as-Commitment).

Sourced to “Time, Derived” (Keeper-PASS v1.3, K1665–K1670). Every scaffold above is labeled a scaffold; the one substantive result is that time is derived.